UNIVERSITY OF SCIENCE
**FACULTY OF INFORMATION TECHNOLOGY**

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# KCPM - 21KTPM2 - MID-TERM

# COURSE: SOFTWARE TESTING

## Requirement 4: Automation testing

*18/08/2024 – 24/08/2024*

**COURSE CODE: CSC13003**
**Group: 11**
**Members:**

| ID | Name |
|---|---|
| 21127699 | Lô Thủy Tiên |

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Trần Duy Hoàng
Trương Phước Lộc
Hồ Tuấn Thanh

**PRESENTER:**
*Lô Thuỷ Tiên*

Ho Chi Minh City – 2024

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# PROJECT INFORMATION

**Course Code:** CSC13003

**Course Name:** Software Testing

**Title:** CS423 – CSC13003 – Software Testing CASE STUDY (PROJECT)

**Topic:** Requirement 4: Automation testing

**Description:**

Software Under Testing: Test Management Website, currently in development with planned releases as follows:

- Prototype: Static web interfaces showcasing the main business functions.
- Build 1: Dynamic system retrieving information from a database, focusing on information display.
- Build 2: Complete system enabling users to manage the software testing process.

**Requirement 4: Automation testing**

Students are asked to use Selenium, Katalon, or other automation testing tools to automate all test cases from the Requirement 2 - Domain testing section. It should be noted that data-driven and checkpoint/assertion/verification techniques are required. Students must use at least 3 Selenium Web Drivers or equivalence for testing on different browsers.

Students must write reports about the automation testing process and the testing results.

In addition, students must record videos of the automation testing process, upload to Youtube and provide the links in the submission.

**Faculty Instructors:** Ms. Trần Thị Bích Hạnh, Mr. Trần Duy Hoàng, Mr. Trương Phước Lộc, Mr. Hồ Tuấn Thanh

# AUTOMATION TESTING REPORT

## I. Introduction:

The purpose of this report is to document the automation testing process for three key features: Users - Register, Projects - Add, and Requirements - Edit. The automation testing was conducted using Katalon Studio, leveraging its Record Web feature to streamline the test creation process. The testing involved executing the test cases developed during Requirement 2 - Domain Testing, and ensuring that data-driven testing and assertion techniques were applied across multiple browsers.

## II. Automation Tool and Techniques:

### 1. *Tool Used: Katalon Studio*

Katalon Studio was chosen for its user-friendly interface and comprehensive support for data-driven testing, as well as built-in assertion libraries.

The Record Web feature was utilized to capture the user interactions required for the test cases. This feature significantly expedited the test case creation process.

### 2. *Testing Techniques:*

Data-Driven Testing: Multiple datasets were used to simulate different user scenarios, ensuring comprehensive coverage of input variations.

Cross-Browser Testing: The test cases were executed on three different browsers: Google Chrome, Mozilla Firefox, and Microsoft Edge. This ensured that the functionalities were consistent across different platforms.

### 3. *Reason for Selection:*

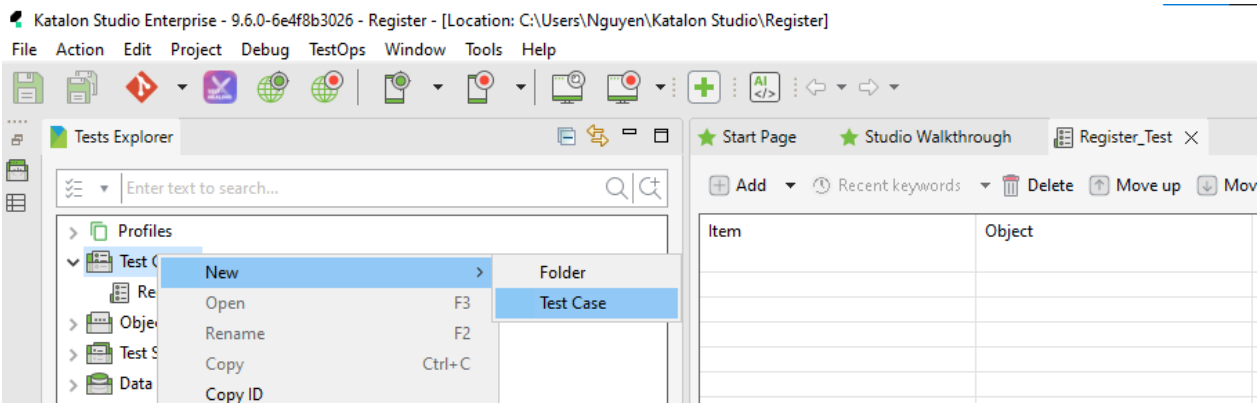
Mockaroo was chosen because it allows for highly customizable field generation and supports a wide range of data types, making it ideal for generating the complex datasets required for this project.

## III. Test Cases Automated

Test case: 21127699-Domain Testing

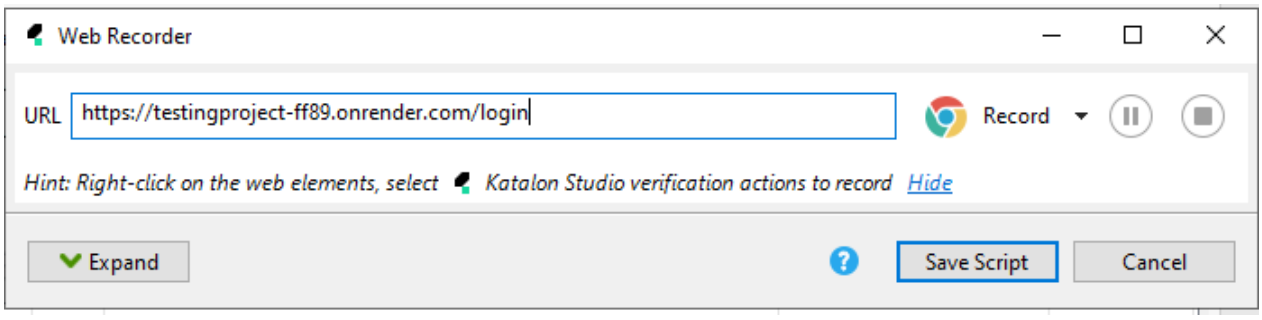
### 1. *Users - Register:*

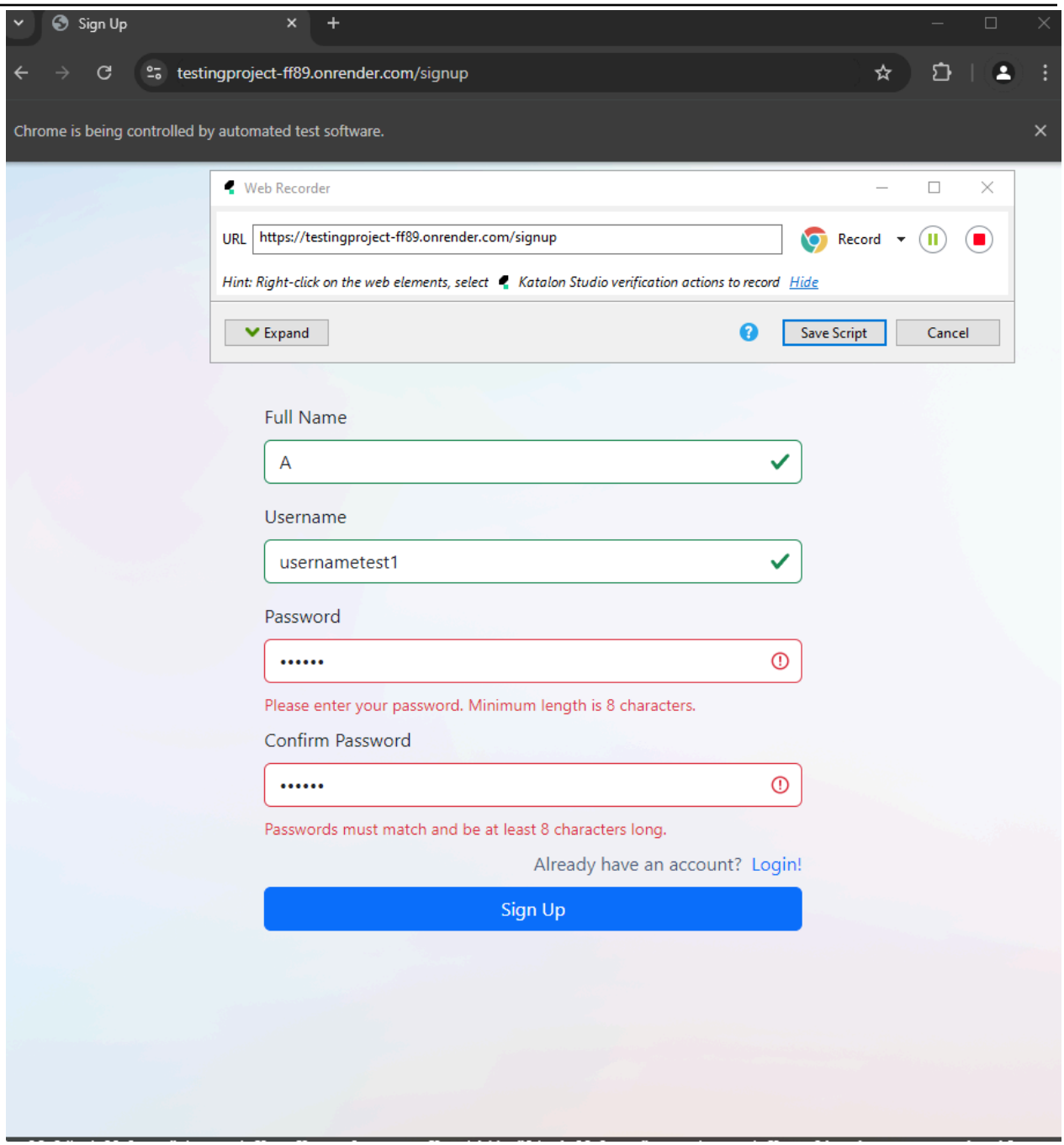
***Step 1: Creating Test Case***




***Step2: Record Web feature. After recording, save the script***

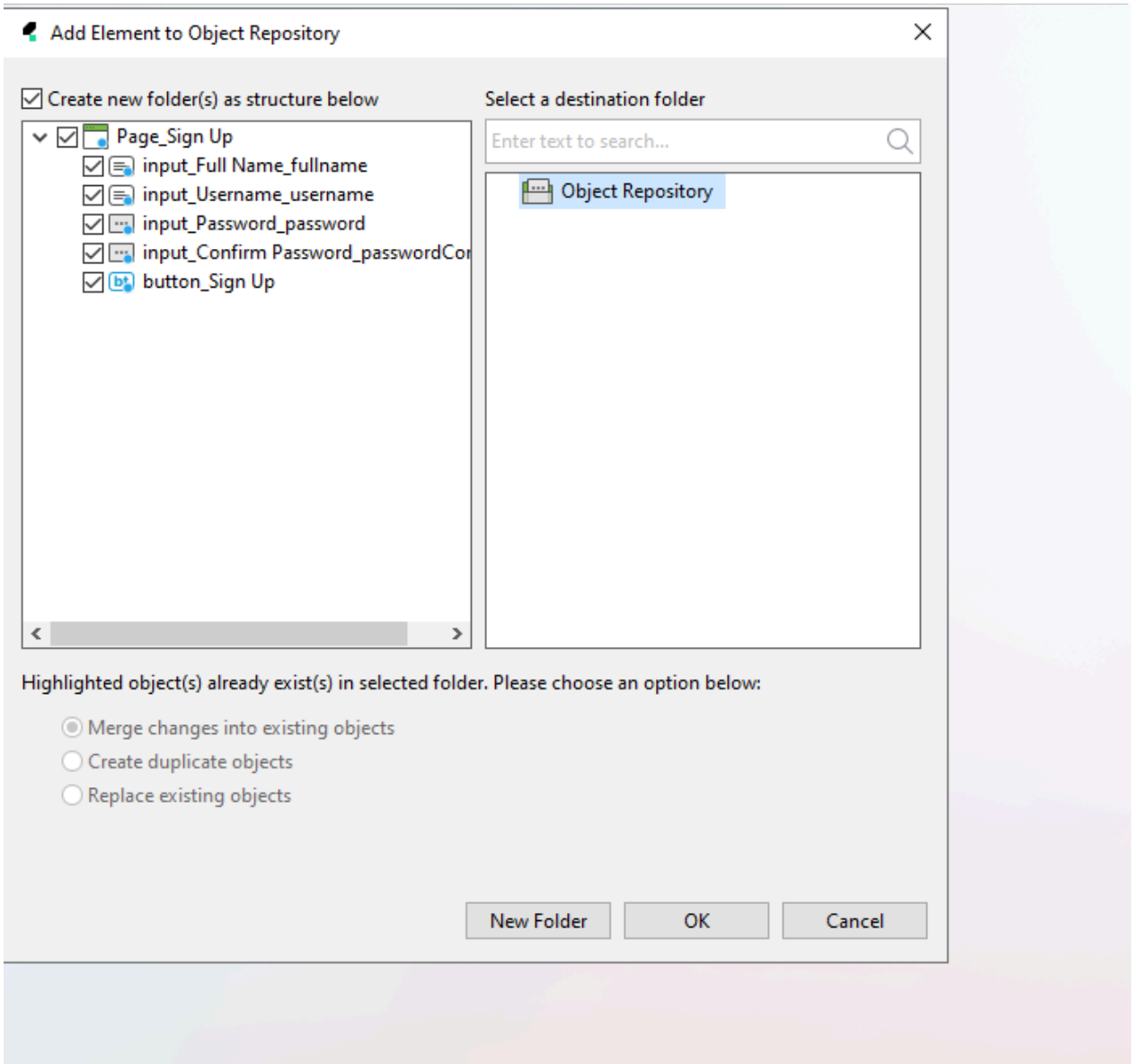
- Used the Record Web feature to capture the registration process, including filling out the fields for Full Name, Username, Password, and Confirm Password.
- Saved the recorded script for further configuration

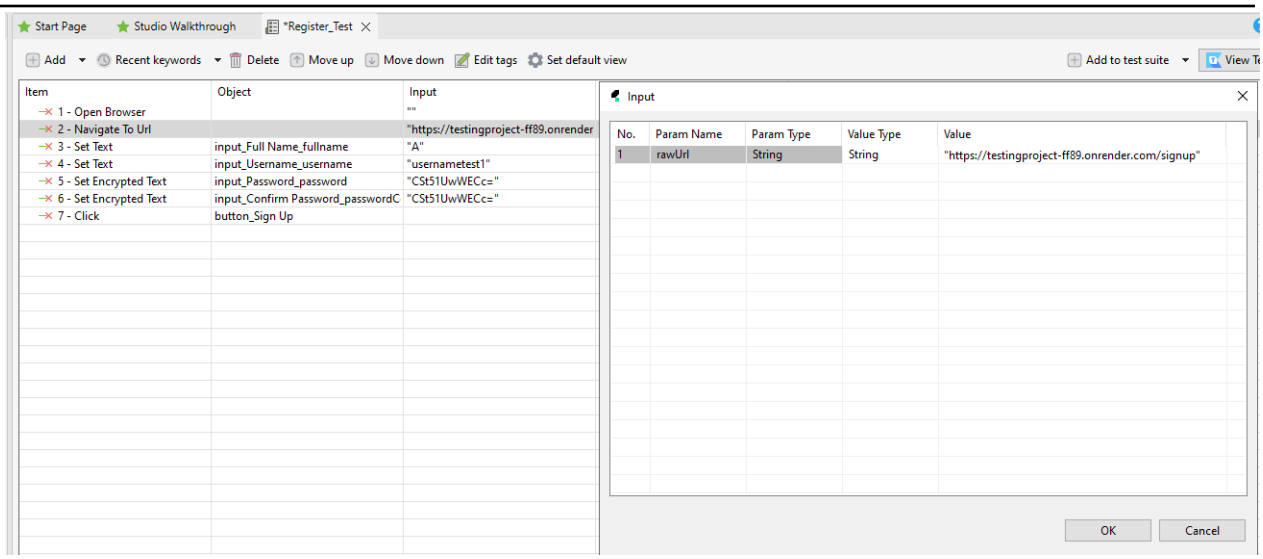


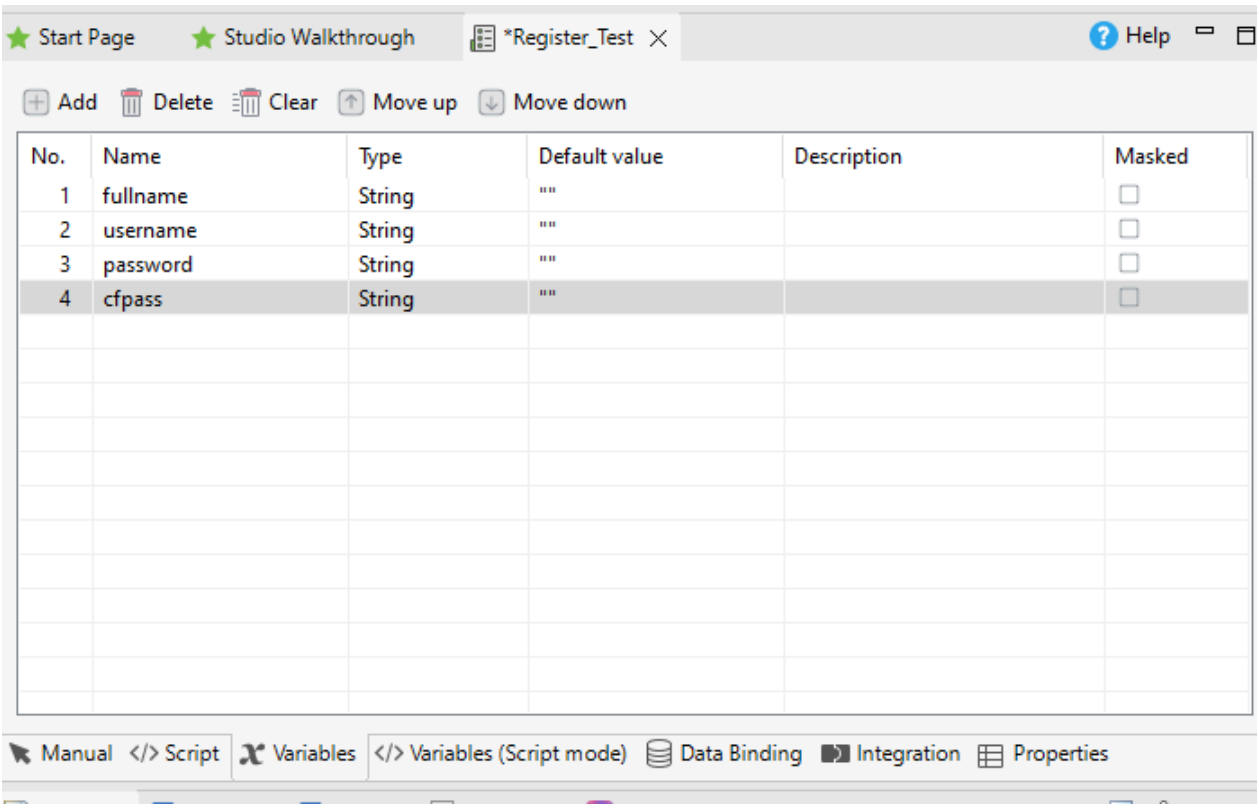


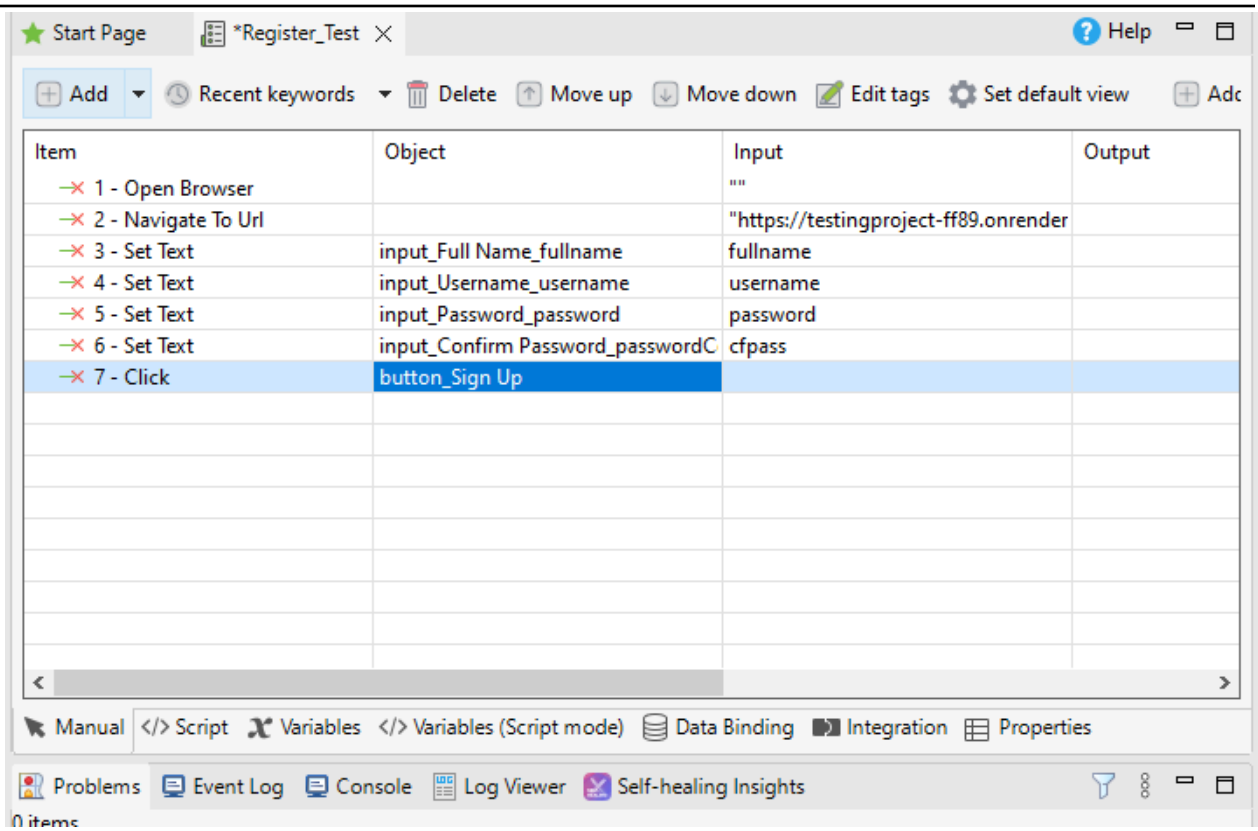

***Step 3: Assign Variables to Test Case***

- Assigned variables to the fields captured during the recording, such as `FullName`, `Username`, `Password`, and `ConfirmPassword`


Add Element to Object Repository
Create new folder(s) as structure below
Select a destination folder
Page_Sign Up
input_Full Name_fullname
input_Username_username
input_Password_password
input_Confirm Password_passwordCor
button_Sign Up
Enter text to search...
Object Repository
Highlighted object(s) already exist(s) in selected folder. Please choose an option below:
Merge changes into existing objects
Create duplicate objects
Replace existing objects
New Folder
OK
Cancel

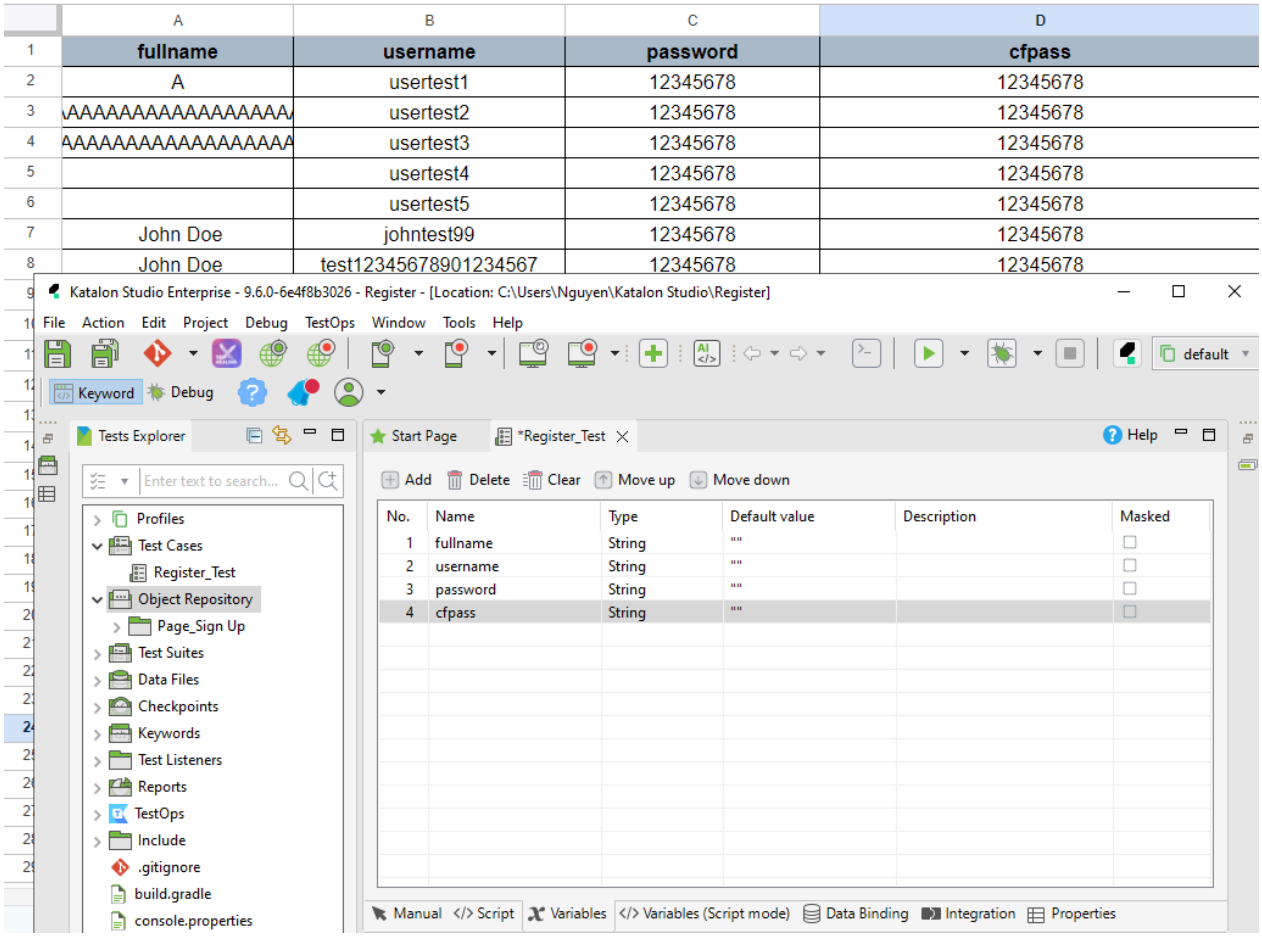

Start Page
Studio Walkthrough
*Register_Test
Add
Recent keywords
Delete
Move up
Move down
Edit tags
Set default view
Add to test suite
Item
Object
Input
1 - Open Browser
2 - Navigate To Url
"https://testingproject-ff89.onrender
3 - Set Text
input_Full Name_fullname
"A"
4 - Set Text
input_Username_username
"usernametest1"
5 - Set Encrypted Text
input_Password_password
"CSt51UwWECc="
6 - Set Encrypted Text
input_Confirm Password_passwordC
"CSt51UwWECc="
7 - Click
button_Sign Up
Input
No.
Param Name
Param Type
Value Type
Value
1
rawUrl
String
String
"https://testingproject-ff89.onrender.com/signup"
OK
Cancel


Start Page
Studio Walkthrough
*Register_Test
Help
Add
Delete
Clear
Move up
Move down
No.
Name
Type
Default value
Description
Masked
1
fullname
String
""
2
username
String
""
3
password
String
""
4
cfpass
String
""
Manual
Script
Variables
Variables (Script mode)
Data Binding
Integration
Properties




***Step 4: Configure Script***

- Edited the script to bind the variables to the input fields.
- Added conditional statements to verify the success or failure of the registration process based on the input data.

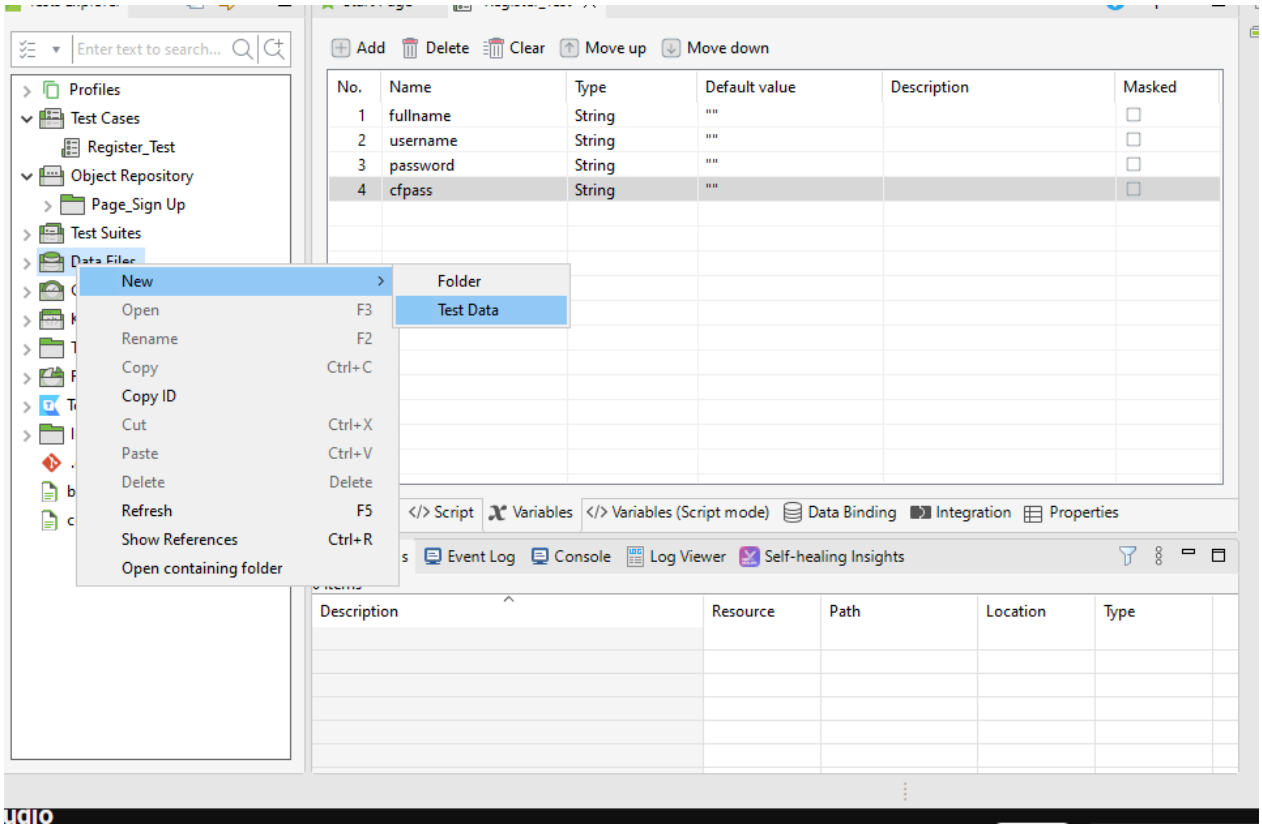


| fullname | username | password | cfpass |
|---|---|---|---|
| A | usertest1 | 12345678 | 12345678 |
| AAAAAAAAAAAAAAAAAA | usertest2 | 12345678 | 12345678 |
| AAAAAAAAAAAAAAAAAA | usertest3 | 12345678 | 12345678 |
| | usertest4 | 12345678 | 12345678 |
| | usertest5 | 12345678 | 12345678 |
| John Doe | johntest99 | 12345678 | 12345678 |
| John Doe | test12345678901234567 | 12345678 | 12345678 |



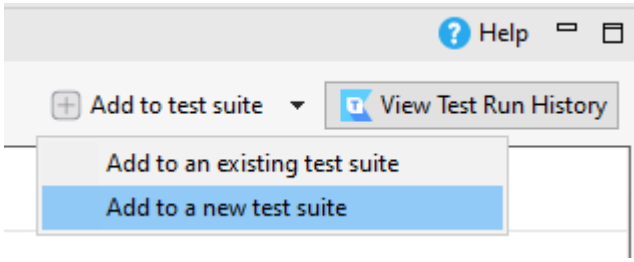
## ***Step 5: Create Data File***

- Created an Excel file containing multiple sets of input data for the registration process.
- Imported the Excel file into Katalon Studio as a data file.




## ***Step 6: Create a Test Suite***

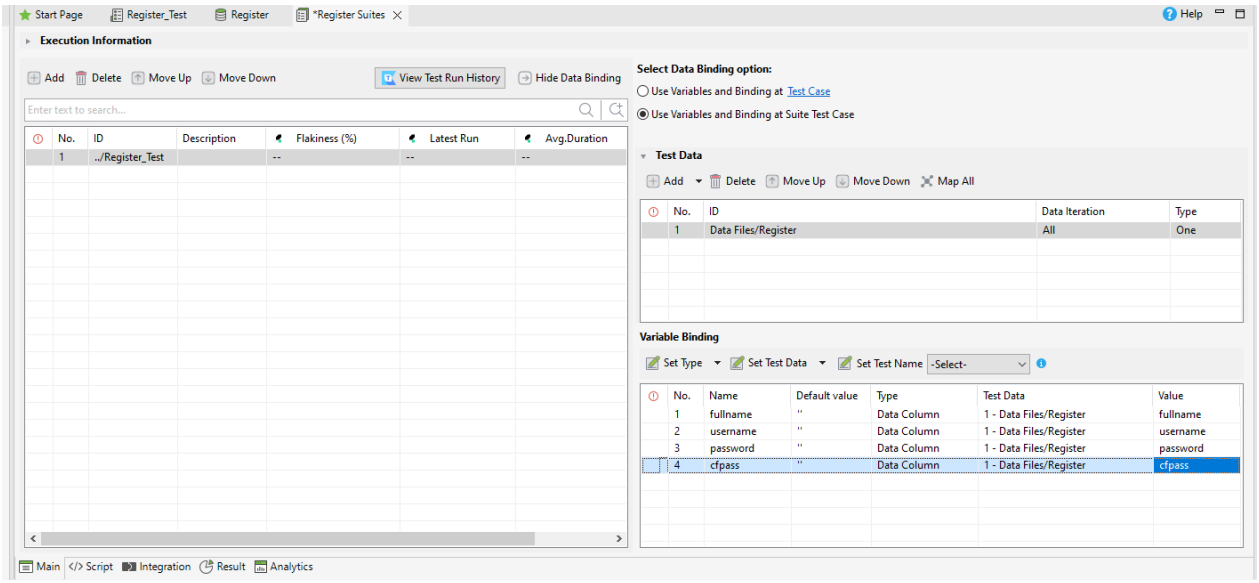
- Created a test suite named RegisterSuite.
- Added the Register test case to the suite and configured it for data-driven testing.




## ***Step 7: Add Test Data and Configure Data Binding***

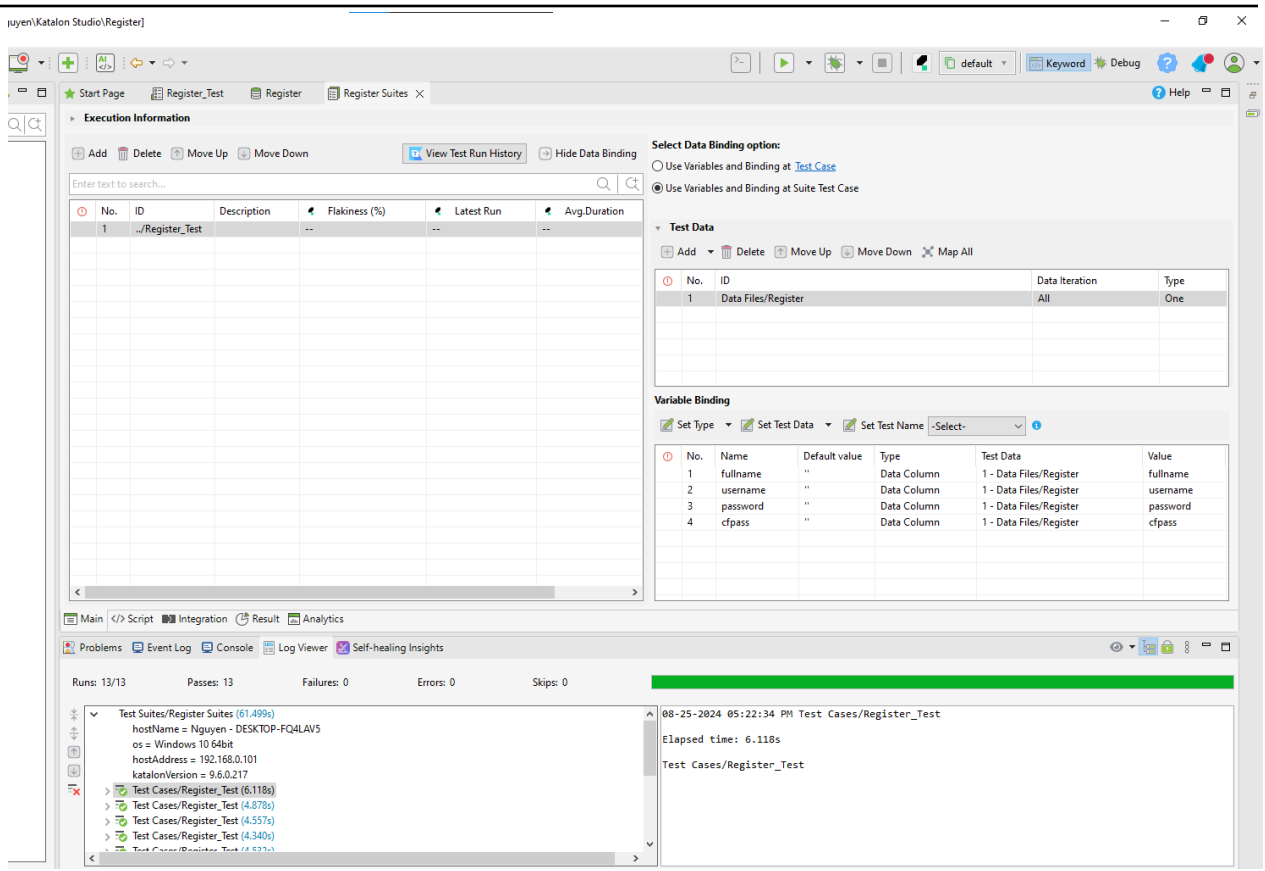
- Added the data file to the RegisterSuite.

- Configured the data binding to match the test case variables with the corresponding columns in the data file



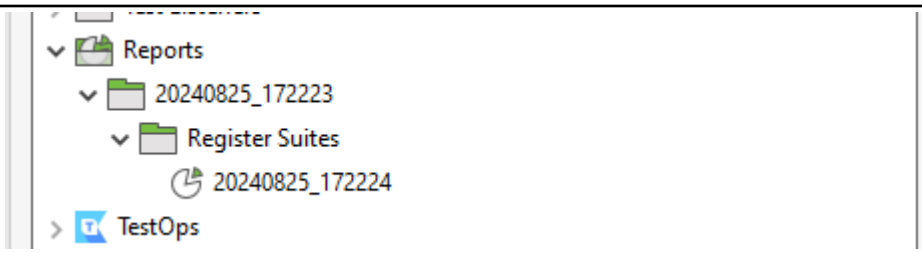

## ***Step 8: Add Test Data and Configure Data Binding***

- Added the data file to the RegisterSuite.
- Configured the data binding to match the test case variables with the corresponding columns in the data file.

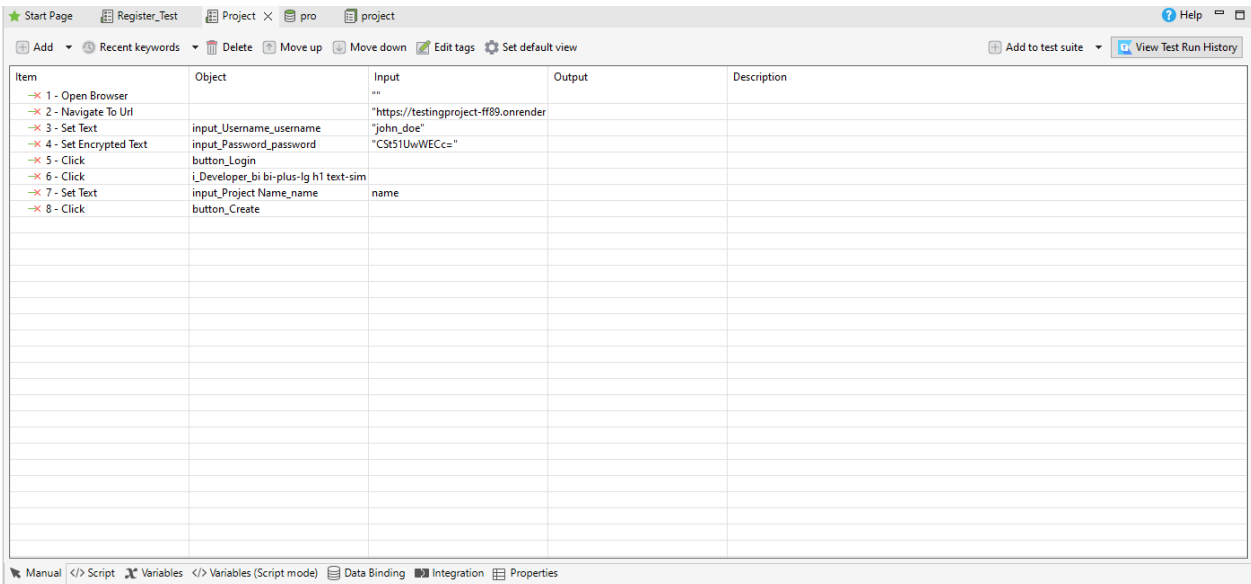



## ***Step 9: View and Analyze Report***

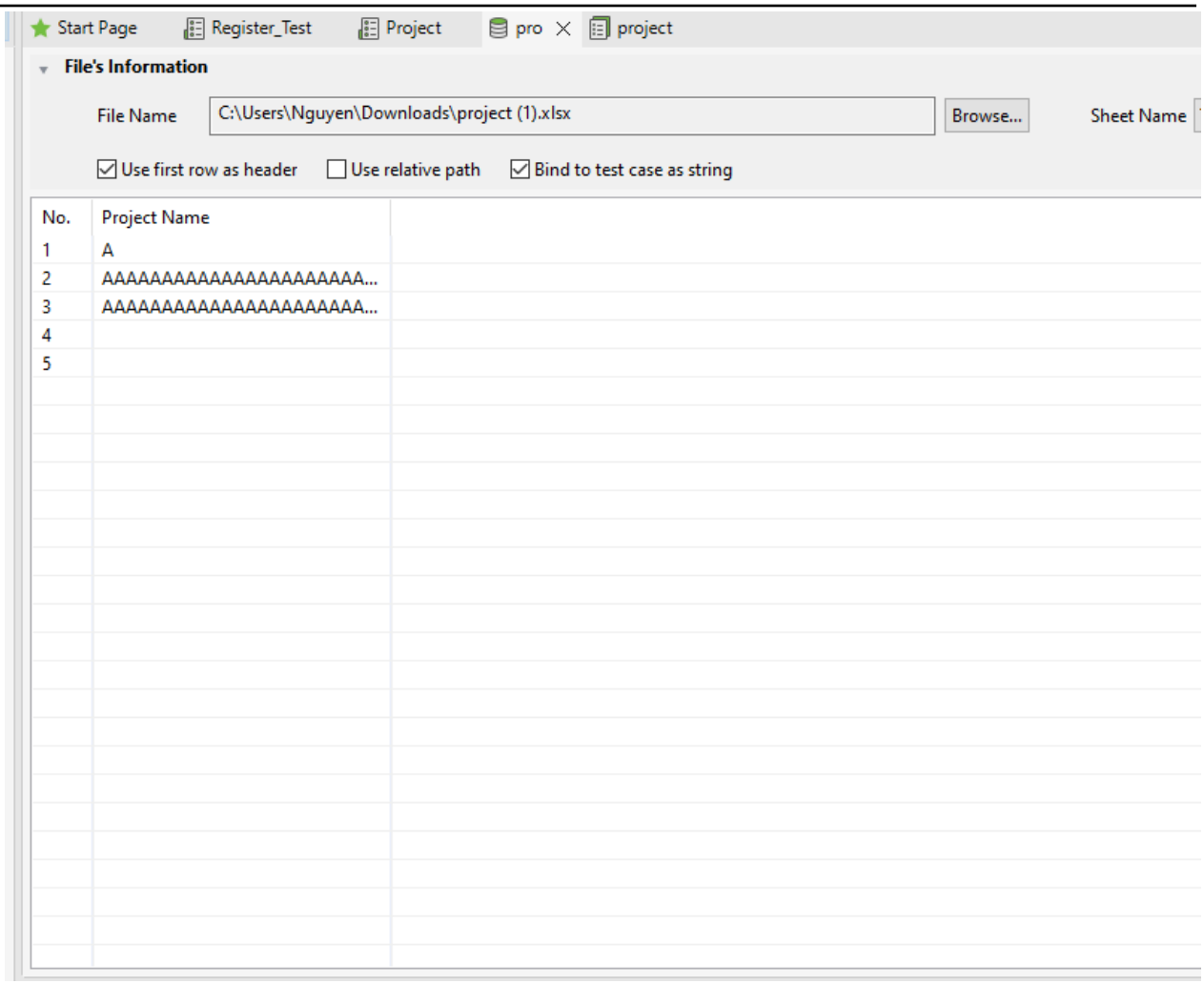
- Ran the test suite collection and viewed the detailed test execution reports.
- Identified and logged any issues encountered during the registration process.

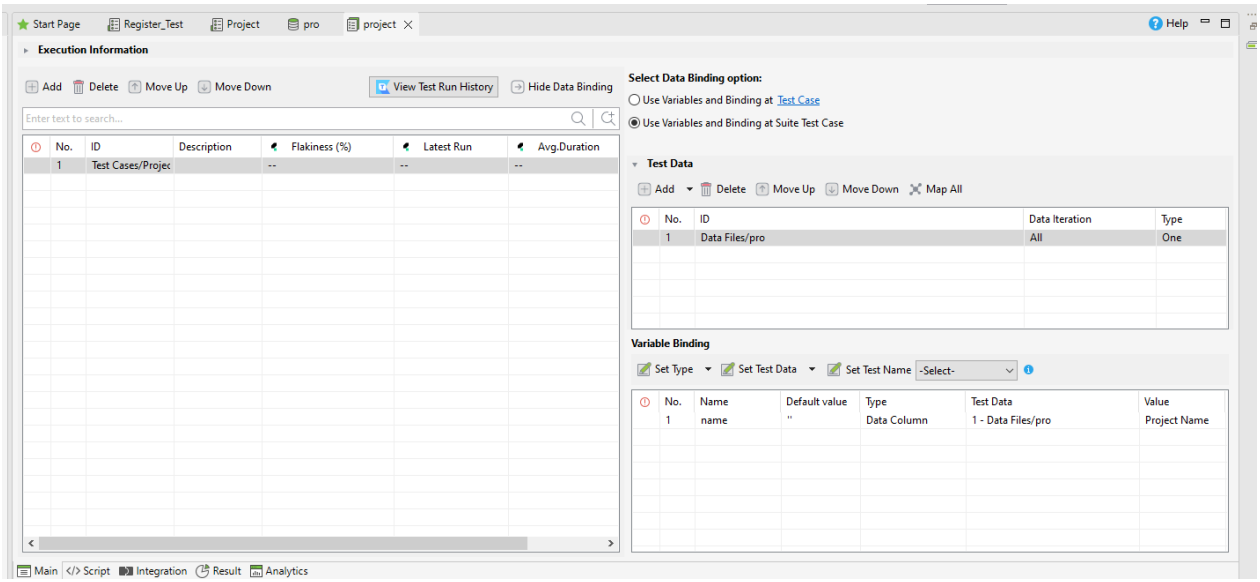



## 2. *Projects - Add:*




| Item | Object | Input | Output | Description |
|---|---|---|---|---|
| 1 - Open Browser | | "" | | |
| 2 - Navigate To Url | | "https://testingproject-ff89.onrender | | |
| 3 - Set Text | input_Username_username | "john_doe" | | |
| 4 - Set Encrypted Text | input_Password_password | "CSt51UwWECc=" | | |
| 5 - Click | button_Login | | | |
| 6 - Click | i_Developer_bi bi-plus-lg h1 text-sim | | | |
| 7 - Set Text | input_Project Name_name | name | | |
| 8 - Click | button_Create | | | |




### 3. *Requirements - Edit*

Start Page | Register_Test | Project | project | Requirement | requirement | Requirement Suites

Add | Recent keywords | Delete | Move up | Move down | Edit tags | Set default view

| Item | Object | Input | Output |
|---|---|---|---|
| 1 - Open Browser | | "" | |
| 2 - Navigate To Url | | "https://testingproject-ff89.onrender | |
| 3 - Set Text | input_Username_username | "john_doe" | |
| 4 - Set Encrypted Text | input_Password_password | "CSt51UwWECc=" | |
| 5 - Click | button_Login | | |
| 6 - Click | div_Renamed ProjectOverviewRequir | | |
| 7 - Click | a_Requirements | | |
| 8 - Click | i_oke_bi bi-pencil-fill text-gray | | |
| 9 - Set Text | input_Requirement Name_name | name | |
| 10 - Set Text | textarea_Requirement Description_de | sp | |
| 11 - Click | button_Edit | | |

File's Information

File Name: C:\Users\Nguyen\Downloads\requiment (2).xlsx | Browse... | Sheet Name: Trang tính4

☑ Use first row as header ☐ Use relative path ☑ Bind to test case as string

| No. | Requirement Name | Requirement Description |
|---|---|---|
| 1 | A | Some description |
| 2 | AAAAAAAAAAAAAAAAAAAAAA... | Some description |
| 3 | AAAAAAAAAAAAAAAAAAAAAA... | Some description |
| 4 | | Some description |
| 5 | | Some description |
| 6 | New Requirement | Some description |
| 7 | New Requirement | |
| 8 | New Requirement | Some description |
| 9 | | |

Execution Information

Add | Delete | Move Up | Move Down | View Test Run History | Hide Data Binding

Enter text to search...

| No. | ID | Description | Flakiness (%) | Latest Run | Avg.Duration |
|---|---|---|---|---|---|
| 1 | ../Requirement | | -- | -- | -- |

Select Data Binding option:

○ Use Variables and Binding at Test Case

◉ Use Variables and Binding at Suite Test Case

Test Data

Add | Delete | Move Up | Move Down | Map All

| No. | ID | Data Iteration | Type |
|---|---|---|---|
| 1 | Data Files/requirement | All | One |
| 2 | Data Files/requirement | All | One |

Variable Binding

Set Type | Set Test Data | Set Test Name -Select-

| No. | Name | Default value | Type | Test Data | Value |
|---|---|---|---|---|---|
| 1 | name | '' | Data Column | 2 - Data Files/requirement | Requirement Na |
| 2 | sp | '' | Data Column | 2 - Data Files/requirement | Requirement De |

## IV. Video Documentation

Link: https://youtu.be/pWUhQ9tPd5o

# REFERENCES

### Supporting Tools and Software:

| No | Function | Tool |
|---|---|---|
| [1] | System | Katalon |
| [2] | Web | Opera GX |
| [3] | Project Management | Discord, Messenger, Google Drive |

### References:

[0]https://www.thedataschool.co.uk/elena-raluca-mazareanu/how-to-in-mockaroo-create-related-datasets-with-a-foreign-key/

[1] https://www.youtube.com/watch?v=S_oYFGhZSkQ

[2] https://www.youtube.com/watch?v=3fGiRwplbLQ